

4 Human breast milk supports the growth and development of a baby.

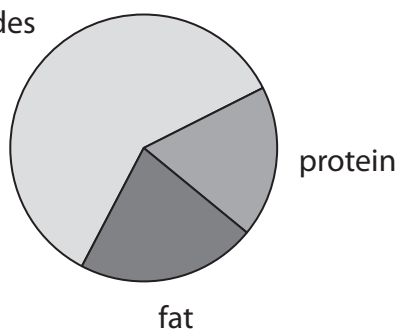
One important component is a group of carbohydrates called oligosaccharides. These contain between three and ten monosaccharide units.

- (a) During the first few days after birth, the milk produced is called colostrum. By two weeks, the composition of the milk has changed and it is called mature breast milk.

The charts show some of the nutritional components of colostrum and mature breast milk.

Colostrum

oligosaccharides



Mature breast milk



Estimate the percentage composition of oligosaccharides in the colostrum.

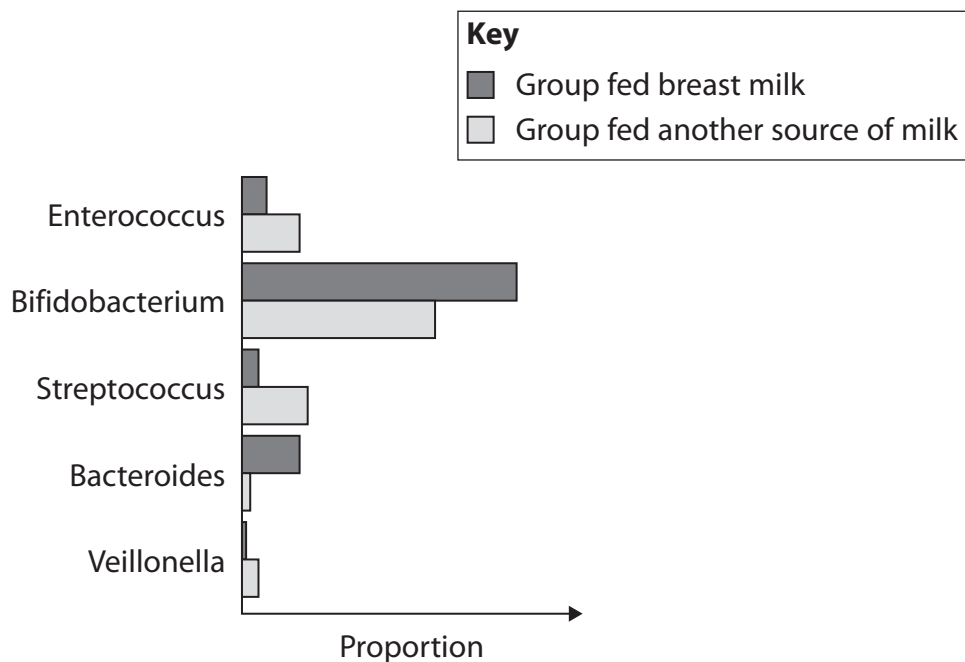
(1)

Answer..... %

(b) Oligosaccharides in breast milk have been shown to be important in establishing a healthy gut flora in the developing baby.

- (i) The effect on the types of bacteria in the guts of babies fed on breast milk or another source of milk was investigated.

The graph shows the proportion of five types of bacteria in the two groups of babies.



Describe **two** conclusions that can be made about the gut flora of babies fed these two types of milk.

(2)

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(ii) Explain the importance of healthy gut flora in preventing infection in a developing baby.

(2)

(c) Group B Streptococcus (GBS) bacteria are a common cause of infections in babies.

These bacteria are becoming increasingly resistant to antibiotics.

Scientists have shown that oligosaccharides isolated from breast milk have antimicrobial activity against GBS bacteria.

Explain the possible advantages of using these oligosaccharides, instead of antibiotics, to treat GBS bacterial infections.

(3)

(Total for Question 4 = 8 marks)